

Question:

If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think> To determine the number of sides of the polygon, let's start by recalling the sum of the interior angles of an (n) -sided polygon, which is given by the formula:

$$(n-2) \times 180^\circ$$

We know that the sum of all the interior angles except one angle is (2190°) . Let the missing angle be (x) . Then, the sum of the remaining $(n-1)$ angles is $((n-2) \times 180^\circ - x = 2190^\circ)$. This can be rearranged to find (x) :

$$x = (n-2) \times 180^\circ - 2190^\circ$$

Since (x) must be a positive angle less than (180°) (as it is an interior angle of a convex polygon), we need to find (n) such that $(0 < (n-2) \times 180^\circ - 2190^\circ < 180^\circ)$. Simplifying the inequality:

$$0 < (n-2) \times 180 - 2190 < 180$$
$$2190 < (n-2) \times 180 < 2190 + 180$$
$$\frac{2190}{180} < n-2 < \frac{2370}{180}$$
$$12.1667 < n-2 < 13.1667$$

Since $(n-2)$ must be an integer, we round 12.1667 up to 13. Therefore:

$$13 < n-2 < 13.1667$$
$$15 < n < 15.1667$$

So, the only integer (n) that satisfies this inequality is $(n = 15)$. Now, let's verify this: The sum of the interior angles of a 15-sided polygon is:

$$(15-2) \times 180^\circ = 13 \times 180^\circ = 2340^\circ$$

If the sum of the remaining 14 angles is (2190°) , then the missing angle (x) is:

$$x = 2340^\circ - 2190^\circ = 150^\circ$$

which is a valid interior angle of a convex polygon. Therefore, the number of sides of the polygon is (15) .

<answer> 15 </answer>